

Matthew Au-Yeung

 matthewway.com |  [M4TTH3](https://github.com/M4TTH3) |  [matthewauyeung](https://www.linkedin.com/in/matthewauyeung) |  matthewauyeung04@gmail.com

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science, Honours Co-op | GPA 3.7

Sept 2022 – Apr 2027

Course: DSA, Operating Systems, Distributed Systems, Concurrency, Compilers, Cryptography & Security

SKILLS

Languages: Java, Kotlin, Python, Rust, Go, JavaScript, TypeScript, C#, Elixir, C/C++

Tools and Frameworks: Spring, gRPC, ASP.NET Core, React, Express.js, net/http, Auth0, Nix

Databases and Cloud: PostgreSQL, Microsoft SQL Server (MSSQL), Firebase, Docker, Github Actions

EXPERIENCE

Software Engineer Intern

May 2025 – Aug 2025

Mechanical Orchard

Remote

- Achieved **60% memory reduction** and **10x throughput** in modernized services by building a **Kotlin FFI API** for Java-style COBOL objects and adding lazy initialization to a **Rust** core
- Reduced mainframe migrations from **weeks to days** by developing a **Python FastMCP** server that extracts program context via **ANTLR** and generates Java templates
- Led the migration of **2 client inventory programs** serving **1K+ stores** using **Java, Spring, Db2, and gRPC**
- Published an open-source Python API (bit.ly/4f9calf) with **4K downloads** to simplify generating Java files

Software Developer Intern

Jan 2025 – Apr 2025

Ford Motor Company

Waterloo, ON

- Built a **Kotlin Android service** capturing CarPlay button telemetry for **5 vehicle-refinement teams**
- Synchronized **4 background services** with **Coroutines** and **Channels** to monitor CarPlay transitions
- Increased CarPlay test coverage from **70% to 83%** using **Coroutines, JUnit, and Robolectric**

Compilers Testing Research Assistant

May 2024 – Jan 2025

University of Waterloo

Waterloo, ON

- Engineered a MIPS assembly emulator and graphical runtime debugger in **Kotlin** used annually by **over 1,000 students** to debug compiler outputs (bit.ly/4gfQ7sh)
- Created an **ANTLR** grammar and parser to collapse bracket-nesting code and minify files from **5 languages**

Software Developer Intern

May 2024 – Aug 2024

Township of Centre Wellington

Fergus, ON

- Shipped a document portal tracking **1K+ locations** using **C# ASP.NET Core, TypeScript, and MSSQL**
- Migrated **Auth0** authorization to middleware-enforced policies, reducing external reliance for **28 municipalities**
- Improved communication for **100+ officials** by building shared notes using **SignalR Websockets** and **Quill.js**

Data Science Intern

May 2023 – Aug 2023

Bio-Ag Consultants and Distributors

Wellesley, ON

- Implemented Python Qt apps to log company metrics, sales, and inventory from **Dynamics365** to Excel, saving **over 3 hours daily** of manual data logging

PROJECTS

Timestamp | *Android, KMP, Spring Boot, Jetpack Compose, PostgreSQL*

m4tth3/timestamp

- Implemented a meetup app serving realtime ETAs using **Jetpack Compose, Kotlin Multiplatform, and Ktor**
- Built a **Spring Boot** backend server integrating **Graphhopper's** routing engine to stream ETAs in **<100ms**

Course Connect | *React Native, FastAPI, PostgreSQL, Auth0*

m4tth3/courseconnect

- Developed a **React Native** app for sharing course content with live chat, secured using Single Ratchet encryption
- Engineered an **Auth0**-secured **FastAPI** backend to centralize CRUD operations with a **PostgreSQL** database

Chess | *C++, React, TypeScript, Express.js, Socket.io, Docker*

matthewway.com/chess

- Architected and developed a **C++** chess game with an MVC and graph-based pattern
- Deployed a **React** interface that communicates via **Socket.io** to the **Express.js** hosted chess engine